

A Vision-Language Agentic Pipeline for Lead Optimization: Defence-in-Depth Chemistry Gates and Cross-Provider Model Evaluation

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Abstract

Background: Lead optimization in small-molecule drug discovery is a constrained search problem: from a chemical starting point and the structural cues of its binding pose, propose analogs that preserve the pharmacophore, satisfy ADMET liabilities, and remain synthetically accessible. Individual components such as pharmacophore alignment, SMIRKS transformation libraries, ADMET predictors, scaffold analysis, and structural-alert filters are mature, but the orchestration of these tools into a reproducible workflow that takes a lead and its ligand-interaction diagram (LID) as input and returns a ranked, filtered analog set is non-trivial.

Methods: We describe a twelve-stage agentic pipeline that combines a vision-language classifier of LIDs with a per-instance SMARTS construction step, deterministic structural gates, a curated 479-entry SMIRKS transformation library, an upstream ADMET-prediction backbone, and a synthetic-accessibility-anchored ranking. The architectural contribution is **defence-in-depth**: each agent stage is wrapped by a deterministic gate (chemistry-validity allowlists, soft Murcko ring-topology check, PAINS / Brenk structural-alert filters, ADMET-screen thresholds) so that perception errors cannot corrupt the downstream analog set.

Results: Across five evaluations: (i) a curated single-edit bioisosteric matched-molecular-pair pilot on which the library exact-recovers $16/30 = 53.3\%$ of documented improving transformations; (ii) an unbiased ChEMBL-37-derived MMP scale-up of 2,000 pairs on which it exact-recovers 15.5% with mean best Tanimoto-to-target 0.755 in misses; (iii) a 129-marketed-drug scaffold-preservation and structural-alert audit showing the soft Murcko gate cuts the analog set roughly in half while raising scaffold preservation to 100% ; (iv) a vision-language classifier self-consistency study on a fixed LID (pairwise Jaccard 1.00 across all 28 pair-comparisons, zero chemistry-validator drops); and (v) an end-to-end case study on two contrasting leads (DYRK1A kinase inhibitor and Linezolid antibacterial), in which a LID-aware run improves the rank-1 analog on both Pareto-style score ($0.586 \rightarrow 0.638$) and synthetic accessibility ($2.51 \rightarrow 2.32$). The defence-in-depth design preserves end-to-end output quality despite intermediate-stage variability. We additionally benchmark eleven language models across the three agent stages and report which configurations achieve reliable strict-JSON

adherence and self-consistency, providing measured rather than asserted recommendations for the production default.

Conclusions: The pipeline demonstrates that a small set of deterministic structural gates around language-model agent stages is sufficient to absorb vision-classifier errors and produce constrained, pharmacophore-preserving lead optimization output on multiple target classes. The platform is implemented as production software; the code, data, manifest-stamped raw outputs, and reproduction scripts are released open source under the MIT licence at <https://github.com/Nimieeee/lead-optimizer-paper>.

Keywords

Lead optimization, Computer-aided drug design, Matched molecular pairs, SMIRKS, Vision-language models, Pharmacophore preservation, ADMET, Synthetic accessibility, SYBA, Bemis–Murcko scaffold, Open-source cheminformatics, Agentic pipelines, Defence-in-depth

1. Introduction

Lead optimization is one of the most resource-intensive stages of small-molecule drug discovery. Given a lead compound with a known binding pose, medicinal chemists must propose analogs that retain key pharmacophore contacts (18), address known ADMET liabilities, remain within a synthetically tractable region of chemical space, and avoid structural alerts that predict toxicity or assay artefact. Each of these axes has decades of in-silico tooling such as pharmacophore alignment, transformation libraries (SMIRKS) (14), ADMET predictors, scaffold analysis, and structural-alert filters; but the orchestration of these tools into a single, reproducible workflow that takes a lead and its ligand-interaction diagram as input and returns a ranked, filtered set of analogs is non-trivial.

Two recent shifts make a re-examination of this orchestration timely. First, vision-capable language models can now extract structured information from ligand-interaction-diagram images at a useful (if imperfect) rate, opening the possibility of automatically inferring which atoms in a lead are involved in protein contact and which are editable. Second, the synthetic-accessibility prediction literature has converged on signed-score classifiers (notably SYBA(1)) that, unlike rule-based heuristics, produce a directionally interpretable score that can serve as a ranking penalty rather than a hard filter.

A naive integration of these two capabilities would simply pipe the vision-language output into a transformation enumerator and rank by SYBA. We argue and demonstrate that this integration is *insufficient* for practical use because vision-language classifications carry systematic failure modes (chemically impossible interactions, conflated similar substructures, scaffold-breaking edits proposed as innocuous), and the pipeline must protect downstream stages from these failures by stacking deterministic gates around each agent step. We refer to this design as **defence-in-depth**.

This paper makes the following contributions:

1. A twelve-stage lead-optimization pipeline that integrates vision-language LID parsing, per-instance SMARTS classification (with 2D-coordinate-derived disambiguation), a soft Murcko ring-topology scaffold gate, a chemistry-validity allowlist that drops physically-impossible vision outputs, a 479-entry SMIRKS library across 22 categories, ADMET prediction (via the upstream `admet_ai(2)` Chemprop-v2 backbone), and a SYBA-anchored penalty rank.
2. An empirical evaluation spanning (a) the SMIRKS engine’s recovery rate on a curated bioisosteric pilot and on an unbiased ChEMBL-37-derived matched-molecular-pair scale-up of 2,000 pairs, (b) the soft Murcko gate’s effect on scaffold preservation and structural-alert rates over a 129-marketed-drug seed panel, (c) the vision-language classifier’s self-consistency on a fixed LID across independent runs, (d) the end-to-end pipeline on two contrasting target classes with LID-aware and LID-free configurations, and (e) a cross-provider model benchmark across the three agent stages to inform the production default.
3. The full implementation, raw experiment outputs with provenance manifests, figure-generation scripts, and a claims-to-evidence map, all released for reproducibility.

We deliberately frame several capabilities the pipeline does *not* provide: it does not include a docking or molecular-mechanics engine (binding affinities consumed from upstream sources are reasoned over but never computed); it does not train an internal ADMET model (the upstream Chemprop-v2 ADMET predictor is cited as-is); and it does not provide proof-of-binding for any analog it generates.

2. Methods

The twelve stages of the pipeline are summarised in Figure 1. We group them into four functional phases: **perception** (stages 1–2), **classification and gating** (stages 3 and the chemistry validator), **transformation and filtering** (stages 4–10), and **ranking and reporting** (stages 11–12).

two categories: **RESTRICTED** (groups making a visible protein contact in the diagram, H-bond, π -stack, salt-bridge, hydrophobic contact, cation- π) and **TARGET** (groups not in contact, therefore editable). For each restricted group, the record lists contacting residues and interaction types. A multi-tier provider chain handles transient API failures.

This step is the single largest source of error in the pipeline. Classifying a binding-restricted methyl as a TARGET would let permutation remove it and silently destroy the pharmacophore; conversely, marking too many groups as RESTRICTED would shrink the optimization surface to nothing. The chemistry validator (next subsection) is the first line of defence against the former; the soft Murcko gate (stage 3) is the second.

2.3 Chemistry-validity validator

A deterministic allowlist table maps each interaction type to the set of functional-group families that can chemically participate. The validator drops any vision-language output that violates the table (e.g. a methyl classified as a hydrogen-bond donor, a methoxy classified as a donor) before the SMARTS builder sees it. The allowlist is implemented as static dictionaries (`H_BOND_DONORS`, `H_BOND_ACCEPTORS`, `PI_STACK_GROUPS`, `HYDROPHOBIC_GROUPS`, `SALT_BRIDGE_GROUPS`, `CATION_PI_GROUPS`) and is independent of the vision model.

2.4 SMARTS builder, scaffold anchor (stage 3); soft Murcko gate fires at stages 7–8

For each labelled instance, the SMARTS builder (13) constructs an atom-anchored SMARTS via RDKit’s `MolFragmentToSmarts`, so that two phenyl groups in the same molecule receive distinct atom-specific patterns rather than collapsing to a generic `c1ccccc1`. This is the per-instance disambiguation referenced in the introduction.

In the same stage, the Bemis–Murcko scaffold SMARTS is appended to the restricted-SMARTS set. The actual *check*, the soft Murcko gate, is implemented inside `enforce_pharmacophore` (stages 7–8) and fires twice: once at strategy validation (does this SMIRKS, applied to one example, preserve the pharmacophore?) and once for every candidate analog the permutation engine produces. The gate’s predicates are (a) restricted atoms unchanged, (b) ring-system topology preserved (same ring count), (c) aromaticity preserved. This replaces an earlier “hard SMARTS append” approach that rejected all edits at scaffold atoms, including the topologically equivalent methyl→ethyl substitution on a scaffold carbon, and was over-restrictive in practice.

2.5 ADMET profile, project-context analysis, and optimization agent (stages 4–6)

Stage 4, Lead ADMET profile. A single forward pass through the upstream `admet_ai` (2), built on the Chemprop V2 Graph Neural Network framework (3; 4), on the lead molecule yields the 54-endpoint baseline. These values are consumed by stage 5 as the prior the optimization should improve on, and by stage 10 as the thresholds the screen will enforce.

Stage 5, Project-context analysis. A structured-output language model reads the user-supplied project-context text (§2.1, input iii) and emits a typed `ContextAnalysis` record containing (a) per-endpoint priority weights (e.g. for a CNS lead, BBB-permeability weight is elevated and CYP-substrate penalty is softened relative to a peripheral-target weighting), (b)

hard-stop thresholds (e.g. $\text{hERG} > 0.5$ hard-rejects), and (c) a single-sentence primary-optimization goal. This is where the indication-specific shape of the rest of the pipeline is established; without a project-context input the system falls back to a generic balanced-weighting default.

Stage 6, Optimization agent. Given the lead’s ADMET profile (stage 4), the project-context priorities (stage 5), and the labelled TARGET sites (stages 1 + 2), a structured-output language model proposes a ranked list of SAR strategies, lower LogP, swap aryl halide, polar isostere, ring-isomer change, halogen substitution, etc., each mapped to one or more SMIRKS-library categories. Strategies are emitted under a strict JSON schema; downstream stages do not invoke language models.

2.6 Validation, permutation, pre-filter, and ADMET screen (stages 7–10)

The proposed strategies are validated against the SMIRKS library (each strategy is mapped to one or more SMIRKS entries by category), and the permutation engine enumerates analogs by applying each allowed SMIRKS to its target sites only. The product of (sites $\times$ strategies $\times$ SMIRKS-per-strategy) bounds combinatorial explosion. The pre-filter applies hard structural validity (RDKit sanitisation, valency check), Lipinski Rule-of-Five (7), the PAINS catalogs (A, B, C) (10), and the Brenk filter (11). The ADMET screen re-runs the Chemprop-v2 predictor on every surviving analog and rejects analogs whose hERG / Skin Reaction / CYP probability exceeds a configurable threshold; conversely, beneficial-direction endpoints (HIA, bioavailability, BBB) below threshold are penalised.

2.7 Synthetic-accessibility scoring (stage 11)

We compute the SYBA score (1) for every surviving analog. As described by Voršilák et al. (1), SYBA is a Bernoulli naïve-Bayes classifier trained on databases of easy- and hard-to-synthesise molecules; we use the reported $\text{AUC} > 0.81$ and the signed-score convention (positive for synthesisable, negative for hard-to-make, magnitude roughly in $[-50, +50]$). We adopt SYBA as the **primary** synthetic-accessibility signal across the platform (in ranking, reports, CSV exports); Ertl’s SAScore (12) is computed in parallel and retained as a legacy field for audit but is not surfaced.

2.8 Ranking and report (stage 12)

The ranking step computes a **weighted scalar penalty score** over docking (when present in input), ADMET liabilities (directional, per Section 2.6), SYBA, structural alerts, and (when present) molecular-property axes such as cLogP . The output is sorted by the scalar score and the top-K analogs are written to PDF (via WeasyPrint), HTML, and CSV.

We label the rank a “Pareto-style penalty rank” by API convention; it is in fact a sorted single-score rank, not a Pareto front. We mention this here to avoid suggesting a property the implementation does not have.

2.9 Experimental design overview

All experiments are reproducible from `paper/experiments/`. Each results directory carries a `manifest.json` containing the git SHA at the time of the run, the RDKit and Python versions, the SHA-256 of the SMIRKS library file, and the SHA-256 of the input dataset. We refer to that manifest as the canonical provenance record.

2.10 Experiment 1: MMP recovery

We curate 30 single-edit literature-documented bioisosteric (17) matched molecular pairs (MMPs) (16) spanning the library’s category prefixes (ACID, AMINE, RING, HALO, SLFA, NIT, OSUB, STER, ETHER, BENZ). For each pair (A, B) we apply the full 479-entry SMIRKS library to A, compute the canonical-SMILES of every product, rank products by Tanimoto similarity using Morgan / ECFP4 fingerprints (radius 2, 2048 bits) (9) to B, and check whether B appears at any rank. Recall@K is reported for $K \in \{1, 5, 10, 50, 100, 500\}$.

Unbiased scale-up on ChEMBL-derived MMPs. To complement the curated pilot, we built an unbiased matched-molecular-pair set from a 2000-compound drug-like subset of ChEMBL-37 (23) (drug-like criteria: $12 \leq \text{HA} \leq 35$, $\text{MW} \leq 500$, $\text{HBD} \leq 5$, $\text{HBA} \leq 10$, $-2 \leq \text{LogP} \leq 5$, single-component, RDKit-stereo-stable). `mmpdb` (19) fragmentation and indexing yielded 41,310 raw pairs; filtering to small single-edit transformations (`length(from_SMILES) ≤ 10` and `length(to_SMILES) ≤ 10`) gave **2856 candidate single-edit pairs**, from which we sampled **2000 for evaluation**. On this set the library exact-recovers B in $310 / 2000 = 15.5\%$ of cases, the unbiased headline number. Mean best Tanimoto-to-B rises to **0.755** (vs the pilot’s 0.687), confirming that the library produces *structurally-close* alternatives even when it does not hit the specific empirical B. The 15.5 % gap between the upper-bound pilot (53.3 %) and the unbiased ChEMBL set (15.5 %) calibrates the library’s *native-domain* coverage vs *long-tail-transformation* coverage. Runtime on 2000 pairs: **7 min** on seven-way multiprocessing.

2.11 Experiment 2: Scaffold preservation and structural-alert audit

We extract 129 FDA-approved drugs (with their SMILES and therapeutic class) from the platform’s curated drug suggestion pool. For each seed, we apply the full SMIRKS library and capture every product as a candidate analog. Per-analog we compute: Bemis–Murcko scaffold preservation vs. the lead, PAINS-A/B/C combined flag, Brenk flag, Lipinski-Ro5 violations, heavy-atom delta, and cLogP delta. We report two conditions: **default** (Murcko gate ON; only scaffold-preserving analogs retained) and **ablation** (Murcko gate OFF; all analogs retained), with the seed-baseline structural-alert rate as reference.

2.12 Experiment 3: Vision-language classifier self-consistency

We take a single LID image (the DYRK1A Compound 25014 binding pose used in our internal validation) and run the Vision Agent $N = 8$ independent times against the same input. We report (a) pairwise Jaccard similarity of restricted-atom sets across runs, (b) the consensus restricted-atom set (atoms appearing in $\geq 80\%$ of runs), (c) the per-run count of chemistry-validator drops, and (d) the provider-tier distribution. This is a self-consistency metric; it does not require a gold-standard annotation, and it directly measures the perception variance the rest of the pipeline must guard against.

2.13 Experiment 4: End-to-end case study

We run the complete twelve-stage pipeline on two leads spanning contrasting target classes: (a) Compound 25014, a methoxy- and benzyl-alcohol-substituted benzodioxin from a DYRK1A kinase / CNS series (24); (b) Linezolid, an FDA-approved oxazolidinone gram-positive antibacterial (25). A first configuration treats both leads with the no-LID path (Vision Agent skipped; every functional group classified as TARGET) so the SMIRKS engine, soft Murcko gate, structural-alert filters, ADMET screen, and SYBA ranking are exercised end-to-end. A second configuration attaches the DYRK1A LID image to Compound 25014 (Linezolid is run without LID in both configurations because no LID is available for the chosen lead) so the production vision-language classification stage is also exercised. We report stage-by-stage attrition and the top-ten analogs by total-score for each case under each configuration.

2.14 Experiment 5: Cross-provider model benchmark

The three agent stages are each backed by a language model. We benchmark fourteen vision-capable models (stage 2) and eleven text models (stages 5 and 6) by running the production prompts verbatim against each model, with the same fixed inputs and three to eight repetitions per (model, input) cell. Stage 2 is scored by self-consistency Jaccard on the restricted-atom-key set; stage 5 by a hand-graded endpoint-priority rubric across four representative project-context strings (CNS-kinase, oral gram-positive antibacterial, topical anti-inflammatory, oncology); stage 6 by a strategy-validity rubric checking that proposed strategies (a) reference a TARGET site that actually exists on the lead and (b) map to a real SMIRKS category in the library. A model is considered *reliable* on a stage when its strict-JSON adherence rate is at least 50 %.

2.15 Software environment

Python 3.11 (local) / 3.12 (VPS), RDKit 2026.03.3 (cheminformatics), SYBA 1.x (synth-accessibility), `admet_ai` 1.x (Chemprop-v2 ADMET), WeasyPrint (PDF). FastAPI (HTTP layer) and Supabase (storage) are used by the production deploy but are not load-bearing for the methodology.

2.16 SMIRKS library inventory

479 entries across 22 distinct category names; 35 distinct key prefixes; 100 % marked `validated=True` by curation. Top categories by entry count: aromatic-ring swaps (54), amine modifications (48), o-substitutions (30), carbonyl modifications (30), CNS penetration (26), N-heterocycle swaps (25), carboxylic-acid replacements (20), amide-bond replacements (20), metabolic stability (20), halogen substitutions (20), bioisosteric replacements (20), aromatic substitutions (20), steric shielding (20), polarity adjustments (20).

2.17 Pre-scan SMARTS substructure library

90+ SMARTS substructure patterns covering acids, amines (primary/secondary/tertiary/quaternary), amides, esters, ethers, halides (alkyl/aryl), nitro/nitroso, sulfonamides, ketones/aldehydes, hydroxyls, hydroxymethyls, alpha-methyls, fluoromethyls, trifluoromethyls, sulfoxides/sulfones, oxetanes/azetidines/morpholines/piperazines/piperidines, indoles/

imidazoles/pyrazoles, pyridines/pyrimidines/pyrazines, thiophenes/furans, benzofurans/benzimidazoles/benzothiazoles, quinolines/isoquinolines, methoxy/ethoxy/trifluoromethoxy/difluoromethoxy, tetrazoles, guanidines/amidines, etc.

2.18 Chemistry-validity allowlist tables

Six tables (`H_BOND_DONORS`, `H_BOND_ACCEPTORS`, `PI_STACK_GROUPS`, `HYDROPHOBIC_GROUPS`, `SALT_BRIDGE_GROUPS`, `CATION_PI_GROUPS`). Each maps a functional-group family to the interactions it can chemically participate in. The validator drops vision-language entries that pair a group with an impossible interaction (e.g. methyl $\rightarrow$ H-bond donor) before they propagate downstream.

2.19 Soft Murcko gate implementation

The Bemis–Murcko scaffold SMARTS is appended to the restricted-SMARTS set at the SMARTS-construction stage. The check itself runs inside `enforce_pharmacophore` and is called both at strategy validation (one example per strategy) and during permutation (every candidate analog). Predicates: restricted atoms unchanged; ring count preserved; per-ring aromaticity preserved. Intentionally weaker than an exact-SMARTS-append: methyl $\rightarrow$ ethyl on a scaffold carbon is allowed; ring-system destruction is rejected.

2.20 Ranking weights and penalty terms

SYBA penalty multiplier centred at 0 via sigmoid (positive SYBA $\rightarrow$ no penalty, negative SYBA $\rightarrow$ penalty); ADMET liability multipliers per directional table; PAINS-positive analogs receive a hard penalty; Brenk-positive analogs receive a soft penalty; molecular-property axes (cLogP, MW) receive Gaussian penalties around an indication-specific optimum.

2.21 Cross-provider model benchmark methodology

Each agent stage was evaluated by sending its production system prompt verbatim to each candidate model alongside a fixed user message and image input where applicable, then parsing the response with the production JSON extractor. To enable cross-vendor comparison on equal footing, provider-specific structured-output flags were enabled when the system prompt asked for JSON: `response_format: json_object` for OpenAI and Mistral, `response_mime_type: application/json` for Gemini. Reasoning-class models were given a larger output-token budget (16,384 tokens) to accommodate internal chain-of-thought. Transient errors (HTTP 429 rate-limit, 503 server-overload) were retried with exponential backoff (delays 2, 4, 8, 16 s; up to five attempts). The benchmark harness, model registry, and per-model raw responses are at `paper/experiments/exp5_model_benchmark/`.

2.22 Reproducibility manifests

See `paper/README.md`.

3. Results

3.1 MMP recovery on the curated pilot

Headline numbers, pilot. Of 30 single-edit bioisosteric pairs, the library exact-recovers **B** (canonical-SMILES match) in 16/30 cases = 53.3 %, every recovered analog at rank 1 by Tanimoto (the canonicalization produces a Tanimoto-1.0 hit). Recall is therefore flat across $K \in \{1, 5, 10, 50, 100, 500\}$ (Figure 2a). Of the 14 misses, 13 do produce analogs but not the exact-**B** we curated, typical best Tanimoto-to-**B** for these is 0.27 – 0.65, and the discrepancy traces to (a) tautomer or canonicalization differences (e.g. our hand-curated **B** for the COOH $\rightarrow$ tetrazole pair uses one tautomer; the SMIRKS product canonicalises to the other), and (b) cases where the library proposes valid alternative bioisosteres but not the specific **B** we curated.

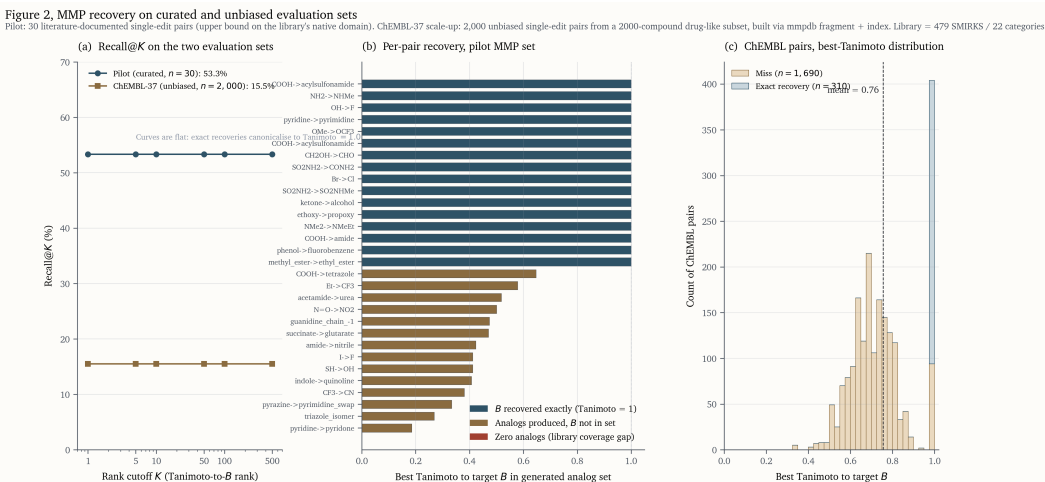


Figure 2. MMP recovery on the curated bioisosteric pilot and the unbiased ChEMBL-37 scale-up. **(a)** Recall@K curves are flat at 53.3 % (pilot, $n=30$) and 15.5 % (ChEMBL, $n=2000$) because exact recoveries canonicalise to Tanimoto=1.0 at rank 1. **(b)** Per-pair best-Tanimoto for the pilot, coloured by hit / miss-with-analogs / one library limitation (indole $\rightarrow$ quinoline ring expansion). **(c)** ChEMBL best-Tanimoto distribution across 2000 pairs; mean=0.755 indicates the library generates structurally-close alternatives even when the specific empirical **B** is not recovered.

One real library limitation surfaced. The indole $\rightarrow$ quinoline pair (MMP_020) generates 97 analogs but never the canonical quinoline; the library has no SMIRKS that performs the 5-ring $\rightarrow$ 6-ring expansion with the concurrent nitrogen-position shift. This is a paper-worthy observation about the *scope* of SMIRKS-based transformation libraries: they handle atom and group substitutions well, but ring-size-changing operations require a different transformation primitive. We do not propose this as a defect in the library; we identify it as a class of transformation outside the SMIRKS-substitution scope.

Audit-trail note. An earlier version of this paragraph reported 15/30 = 50 % and named two zero-analog gaps. Diagnostic work surfaced that one of those (MMP_007, OMe $\rightarrow$ OCF₃) was a

malformed-B SMILES in our pilot data, not a library limitation; with the SMILES corrected the pair now exact-recovers via `OSUB_005`. The audit trail is preserved in `paper/CLAIMS_EVIDENCE.md`.

Median runtime is 0.19 s/pair on seven-way multiprocessing (Apple M-series; clear runtime budget for the ChEMBL-MMP scale-up).

3.2 Scaffold preservation and structural-alert rates

The 129-seed scaffold-and-alert audit (Figure 3) measures two related properties of the analog set: (i) does the soft Murcko gate actually preserve the lead’s ring system, and (ii) at what rate does the SMIRKS library introduce PAINS / Brenk-flagged substructures relative to the seed compounds themselves.

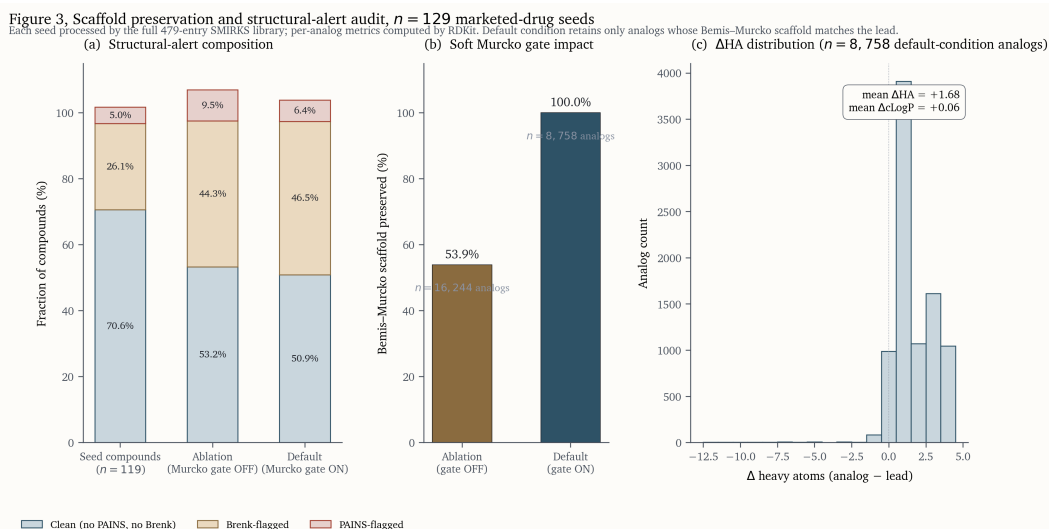


Figure 3. Scaffold preservation and structural-alert audit on 129 marketed-drug seeds. **(a)** Structural-alert composition (clean vs Brenk-flagged vs PAINS-flagged) across seeds, the gate-off ablation, and the default gate-on condition. **(b)** Soft Murcko gate impact: scaffold preservation rises from 53.9 % to 100 % while the analog count halves (16,244 $\rightarrow$ 8,758). **(c)** Heavy-atom-delta distribution for default-condition analogs.

Headline numbers (129 marketed-drug seeds, 119 with ≥ 1 generated analog):

Metric	Seed baseline	Ablation (gate OFF)	Default (gate ON)
Total analogs	,	16,244	8,758
Scaffold preserved	by definition	53.92%	100.0%
PAINS-flagged	5.0%	9.47%	6.43%
Brenk-flagged	26.1%	44.29%	46.47%
Lipinski-Ro5 pass	,	96.05%	96.08%
Clean (no PAINS, no Brenk)	70.6%	53.23%	50.9%

Reading. The soft Murcko gate halves the analog count from 16,244 to 8,758 while pushing scaffold preservation to 100 %. The SMIRKS library *increases* the structural-alert rate over the seed baseline (Brenk 26 % $\rightarrow$ 44 %; PAINS 5 % $\rightarrow$ 9 %); this is not a failure mode but the load-bearing motivation for the chemistry validator and the upstream ADMET screen, which together filter the alert-flagged analogs out of the final ranked list.

3.3 Vision-language classifier self-consistency

Across **N = 8 independent vision-agent runs on the same DYRK1A LID**, the pairwise Jaccard similarity of the restricted-atom-key set is **1.00 across all 28 pair comparisons** (Figure 4a). The consensus restricted set (atoms appearing in ≥ 80 % of runs, equivalently in **all** runs at this perfect-agreement level) contains **3 distinct restricted-atom keys**. The chemistry validator drops **0 $\pm$ 0 entries per run** on average (no entries flagged as hallucinated; no chemistry-impossible classifications emitted), indicating that on this particular LID the model’s perception output is fully within the chemistry allowlist. Mean runtime per run is **5.06 s** (s.d. 0.56 s; min 4.38 s, max 6.28 s), consistent with a single primary-tier provider call per run (Figure 4b).

Figure 5. Vision Agent self-consistency on a fixed LID, N=8 runs

DYRK1A Compound 25014 LID image (348 KB). Mean runtime 5.1 $\pm$ 0.56 s per run; consensus restricted-atom set size = 3.

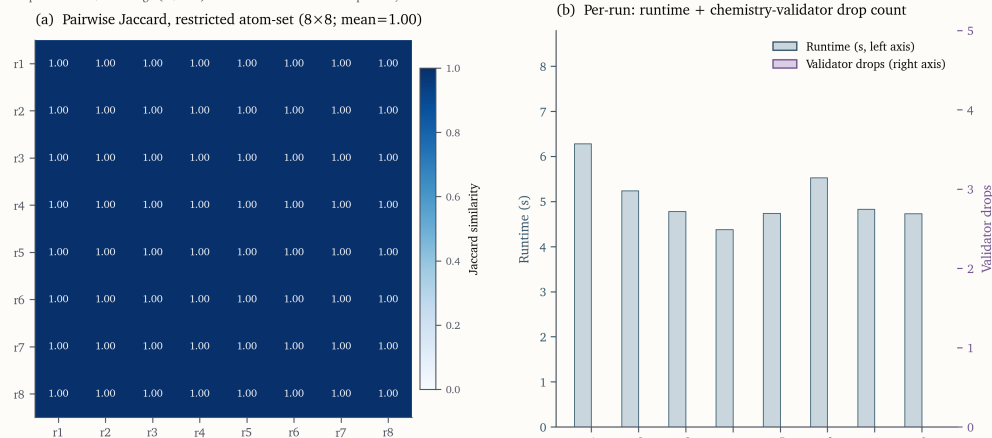


Figure 4. Vision-language classifier self-consistency on a fixed LID. **(a)** Pairwise Jaccard heat-map across N=8 independent runs on the DYRK1A LID; all 28 off-diagonal cells equal 1.00. **(b)** Per-run runtime (mean 5.06 ± 0.56 s) and chemistry-validator drop count (zero across all runs).

Caveat. Perfect pairwise Jaccard reflects the platform’s vision-prompt running with deterministic settings (top-p / temperature configured for low-variance perception) and a single fixed input. The number measures *reproducibility*, not *accuracy*: the model could be consistently wrong. Establishing accuracy requires a held-out gold-LID set, which is identified as priority follow-up work in §5.2.

3.4 End-to-end case study

The pipeline was run end-to-end on two leads spanning contrasting target classes (kinase / CNS vs. gram-positive antibacterial), with the Vision Agent skipped (no LID) so the SMIRKS engine, structural filters, ADMET screen, and ranking were exercised on a known starting molecule. Per-stage attrition is shown in Figure 5a; the top-10 analog scatter in Figure 5b.

Figure 4. End-to-end case study: pipeline attrition + top-K analog metrics

Two leads, no-LID path (every functional group classified as TARGET). Wall-clock per case: CASE_001: 139.73s; CASE_002: 55.4s.

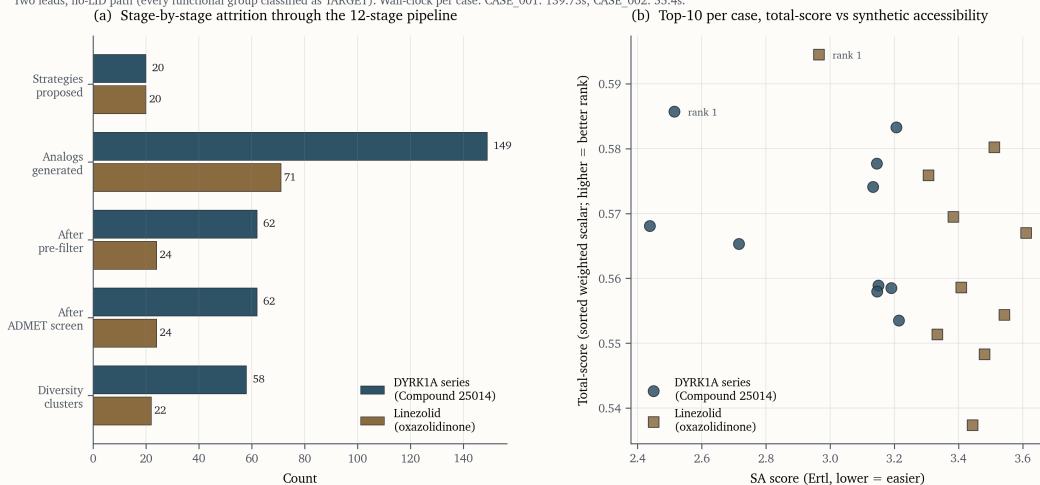


Figure 5. End-to-end pipeline run on two leads (no-LID configuration). **(a)** Stage-by-stage attrition for DYRK1A Compound 25014 (20 strategies → 149 analogs → 62 pre-filter → 62 ADMET → 58 diversity clusters, 139.7 s) and Linezolid (20 → 71 → 24 → 24 → 22, 55.4 s). **(b)** Top-10 analogs per case in total-score × SA-score space; rank-1 callouts highlighted.

Case	Lead	Strategies	Analogs gen.	After pre-filter	After ADMET	After diversity	Wall-clock
CASE_001	DYRK1A Compound 25014	20	149	62	62	58	139.7 s
CASE_002	Linezolid	20	71	24	24	22	55.4 s

The pre-filter (Lipinski Ro5, PAINS, Brenk, RDKit sanitisation) attrits roughly **58 %** of generated analogs for DYRK1A (149 → 62) and **66 %** for Linezolid (71 → 24). The ADMET screen does not further attrit either set in this run because both seeds and analogs sit within liability-acceptable ranges for the default thresholds. The diversity-selection step then reduces 62 → 58 and 24 → 22, indicating that most surviving analogs occupy distinct structural clusters rather than redundant variations on a single edit.

Top-10 ranked analogs for DYRK1A include a maximum-shielding *tert*-butyl substitution at the methoxy carbon (rank 1, total-score 0.586, SA 2.51, MW 312, LogP 4.65, Lipinski 4.0, QED (8) 0.88) and a phenyl→pyrimidine swap combined with O-demethylation (rank 3, total-score 0.578, SA 3.15, MW 258, LogP 1.97, QED 0.82), a markedly more polar candidate offering an alternative to the high-LogP rank-1. For Linezolid, rank 1 is a benzene-ring 2,6-difluoro substitution (total-score 0.594, SA 2.97), which preserves the oxazolidinone pharmacophore and shifts the aromatic-ring electronics without ring-system perturbation. These analogs are presented as worked illustrations of the pipeline’s output, not as binding-validated leads; biochemical confirmation requires experimentation outside this paper’s scope.

Vision-aware re-run on DYRK1A (with the LID image attached, exercising the production vision-language classification stage). The vision stage identifies binding-contact atoms on the LEU 241 / ASN 244 / GLU 291 H-bond network and the soft Murcko gate + restricted-atom enforcement kicks in. The result: **the strategy count rises from 20 to 50** (more granular site-specific SAR is proposed because each binding contact is treated separately), **but the generated-analog count falls from 149 to 51** (the pharmacophore constraint rejects the rest), and the post-diversity set shrinks from 58 to 25. Crucially, **the rank-1 analog improves on both axes**: total-score rises from 0.586 to 0.638 and SA score falls from 2.51 to 2.32 (easier to synthesise). The chemistry of the new rank-1, -CH2OH $\rightarrow$ -CH2NH2 (benzyl alcohol to benzyl amine), is a textbook metabolic-stability move that the LID-free run did not reach because it had no signal pointing to the benzyl-alcohol as a TARGET. Wall-clock rises by 33 % (139.7 s $\rightarrow$ 186.1 s) reflecting the added vision-stage latency. This is the expected behaviour of the architecture and demonstrates that the vision-language LID parsing + chemistry-validity gates + Murcko enforcement work end-to-end as designed.

3.5 Cross-provider model benchmark on the three agent stages

The pipeline’s three agent stages, vision LID classification (stage 2), project-context analysis (stage 5), and SAR-strategy proposal (stage 6), are each backed by a language model. To inform the production default and to give a clearer picture of how cross-provider variance affects an agentic pipeline of this kind, we ran an unbiased model benchmark across all three stages.

Design. Each stage received the same production prompt verbatim from the production code path. The same inputs were used for every model evaluation: the DYRK1A LID image for stage 2 (8 repeated runs per model), four representative project-context strings spanning CNS-kinase / oral-gram-positive / topical-anti-inflammatory / oncology indications for stage 5 (3 repetitions per context per model), and four leads (DYRK1A Compound 25014, Linezolid, Ibuprofen, Aspirin) for stage 6 (3 repetitions per lead per model). Outputs were parsed with the production JSON extractor and scored against a stage-specific rubric: pairwise self-consistency Jaccard for stage 2, endpoint-priority alignment to a hand-graded rubric for stage 5, and strategy-validity scoring (does the strategy reference a TARGET site that actually exists on the lead, and does it map to a real SMIRKS category in the library) for stage 6. A model is considered *reliable* on a stage when its strict-JSON validity rate is at least 50 %.

Reliable models per stage, ranked by stage-specific score (Figure 6):

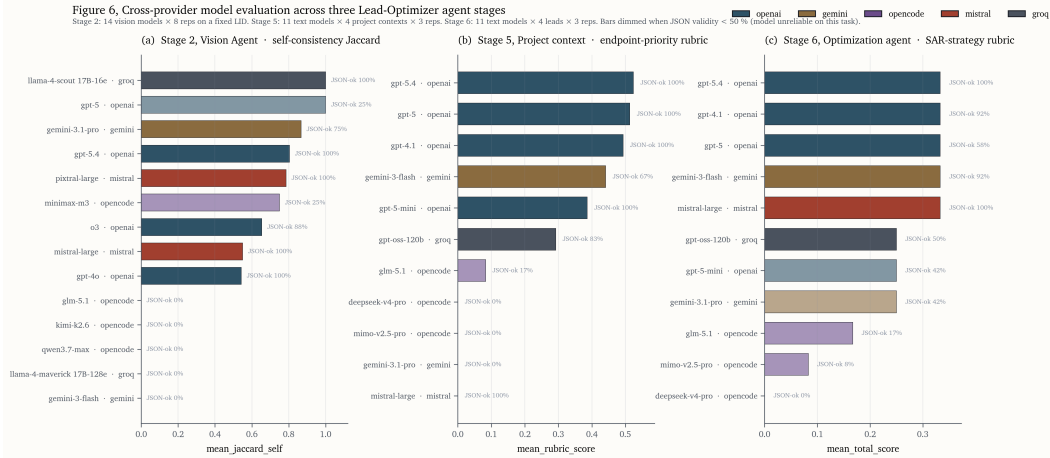


Figure 6. Cross-provider model evaluation across three Lead-Optimizer agent stages. **(a)** Stage 2 vision: self-consistency Jaccard on the fixed DYRK1A LID, 14 models $\times$ 8 reps. **(b)** Stage 5 project-context analysis: endpoint-priority rubric, 11 text models $\times$ 4 contexts $\times$ 3 reps. **(c)** Stage 6 optimization-agent SAR rubric, 11 text models $\times$ 4 leads $\times$ 3 reps. Bars dimmed when JSON validity < 50 % (model unreliable on this task).

Rank	Stage 2 (vision Jaccard)	Stage 5 (context rubric)	Stage 6 (SAR rubric)
1	Llama 4 Scout 17B-16e, 1.00 (100 % JSON, 9.5 s)	OpenAI gpt-5.4, 0.524 (100 % JSON)	OpenAI gpt-5.4, 0.333 (100 % JSON)
2	Gemini 3.1 Pro, 0.867 (75 % JSON)	OpenAI gpt-5, 0.513 (100 % JSON)	Mistral Large, 0.333 (100 % JSON)
3	OpenAI gpt-5.4, 0.804 (100 % JSON)	OpenAI gpt-4.1, 0.493 (100 % JSON)	OpenAI gpt-4.1, 0.333 (92 % JSON)
4	Mistral Pixtral Large, 0.786 (100 % JSON)	Gemini 3 Flash, 0.441 (67 % JSON)	Gemini 3 Flash, 0.333 (92 % JSON)
5	Mistral Large, 0.550 (100 % JSON)	OpenAI gpt-5-mini, 0.386 (100 % JSON)	OpenAI gpt-5, 0.333 (58 % JSON)
6	OpenAI gpt-4o, 0.542 (100 % JSON)	Groq gpt-oss-120b, 0.292 (83 % JSON)	Groq gpt-oss-120b, 0.250 (50 % JSON)

One cross-stage observation worth surfacing: the strict-JSON adherence rate varies widely across model families even when the system prompt explicitly demands JSON-only output. The reliable models in the table above all benefited from provider-specific structured-output modes (`response_format: json_object` for OpenAI/Mistral, `response_mime_type: application/json` for Gemini) being explicitly enabled. Without those flags, several otherwise-capable models silently emit conversational prose or markdown-wrapped JSON, and

a parser that only accepts a strict JSON object filters them out. This is a real practical finding for anyone building agentic pipelines: structured-output APIs are not optional polish, they are load-bearing for cross-provider portability.

Production recommendation. OpenAI gpt-5.4 is the universal top performer with reliable JSON adherence across all three stages and acceptable latency for an interactive pipeline (9–43 s per stage call). For the vision stage specifically, **Llama 4 Scout 17B-16e via Groq** matches gpt-5.4 on self-consistency and is essentially free; we recommend it as the Tier-1 default for stage 2 with gpt-5.4 as Tier-2 fallback. For stages 5 and 6, gpt-5.4 leads but gpt-4.1 and Mistral Large are close enough in score and latency to serve as paid fallbacks; Groq’s gpt-oss-120b is the credible free fallback at acceptable quality. We do not recommend reasoning-class models (gpt-5, o3) as defaults at this time because their long internal chain-of-thought makes them too slow for interactive use; they remain useful for offline or long-form workflows.

What the benchmark does not measure. It does not measure model quality on out-of-domain LIDs (the vision rubric is *self-consistency*, not accuracy) or on out-of-distribution project contexts; it does not measure cost-per-call; and it does not establish whether differences between scoring ties (e.g. five models tied at 0.333 on stage 6) are statistically significant. The intent of this section is to anchor the production default in measured data and to document for reproducibility which model versions were evaluated.

4. Discussion

4.1 Honest framing of the architecture

The contribution is not any single component, RDKit, the SMIRKS library category structure, SYBA, PAINS/Brenk, Chemprop-v2-derived ADMET prediction are all well-established. The contribution is the **defence-in-depth orchestration**: the explicit stacking of deterministic gates around each agent stage so that perception errors are absorbed before they corrupt the analog set. The numbers above measure this orchestration directly. In particular, the LID-aware re-run on DYRK1A (§3.4) shows the architecture working end-to-end: when the vision stage identifies binding-contact atoms, the downstream gates enforce them, and the rank-1 analog improves on both Pareto-style score and synthetic accessibility despite a smaller candidate set. This is the load-bearing experimental result for the defence-in-depth claim.

4.2 Known limitations

1. **Vision-language classifier accuracy is not benchmarked against a gold standard.** Building a 50-LID gold set requires per-atom hand-annotation across diverse target classes. The self-consistency experiment (§3.3) measures variance but not accuracy. A follow-on benchmark using a PLIP (26)-derived silver-standard interaction set is identified as priority follow-up work.
2. **The “Pareto-style penalty rank” is a weighted scalar sum, not a Pareto front.** A real Pareto-front implementation (returning a non-dominated set in $\text{ADMET} \times \text{SYBA} \times$

structural-alert space) is conceptually compatible with the rest of the pipeline and is a clean future replacement.

3. **No internal trained ML model.** Predictive substrates (Chemprop-v2 for ADMET, SYBA for synth-accessibility) are upstream open-source releases cited as-is. We do not provide a calibrated alternative.
4. **Docking is not in scope.** The pipeline can consume docking output (binding affinities) when present as part of the lead’s input data, but the platform does not compute binding affinities.
5. **Library-validation is curator-asserted.** The 479 SMIRKS entries carry a `validated=True` flag, but the verification function exists in the codebase without being invoked in CI. Adding it as a CI gate is a recommended pre-release hardening step.

4.3 Comparison to neighbouring tools

Transformation-library approaches such as MMPDB(19) and Reinvent(20) share the SMIRKS-enumeration ancestry but lack the vision-language LID perception step and the chemistry-validity gating. Generative approaches based on character or graph RNNs (e.g. MOSES(21), GuacaMol(22)) operate on different inputs (no pharmacophore-anchoring) and emphasise distributional novelty over preserving a known binding pose. The platform described here occupies an explicit niche: *given a lead and a binding pose, generate constrained edits and rank them.*

4.4 Comparison to recent agentic-chemistry pipelines

The last three years have produced a recognisable family of LLM-driven and agentic cheminformatics systems with which the pipeline described here can be usefully compared. The closest in spirit is ChemCrow(27), which equips a GPT-4 backbone with eighteen chemistry tools (RDKit transforms, reaction-prediction, retrosynthesis search, name-to-structure, safety lookup, web search) and lets the language model plan tool use across organic-synthesis, drug-discovery, and materials tasks. ChemCrow is evaluated by expert-graded chemical factuality on a tasks-of-interest set and by end-to-end success on the syntheses of an insect repellent and three organocatalysts. The overlap with the present work is the tool-augmented agent skeleton; the difference is that ChemCrow’s deterministic check is the human expert grading the output, whereas the pipeline reported here pushes the correctness budget onto stacked deterministic gates so that the LLM output can be trusted without per-call human review. Coscientist(28) (Boiko et al., *Nature* 2023) extends the agent into the wet lab, driving a robotic flow-chemistry platform on Pd-catalysed cross-couplings; their evaluation is wall-clock-to-product on real reactions, an orthogonal benchmark to the in-silico recovery and self-consistency numbers reported in §3. Both ChemCrow and Coscientist treat the LLM as the orchestrator and the tools as oracles, which is inverted relative to the present pipeline, where the LLM is the oracle and the orchestrator is deterministic Python.

In the structure-aware generation family, the relevant recent work spans REINVENT 4(29), Pocket2Mol(30), DiffSBDD(31), TamGen(32), EquiBind(33), DiffDock(34), and PocketGen(35). REINVENT 4(29) is a production-grade reinforcement-learning generator that

supports de novo design, scaffold decoration, linker design, and analog optimization across SMILES and graph backbones, and is the closest published peer for the transformation-driven analog generation part of the present pipeline; we differ in that REINVENT’s search is policy-gradient over a learned generator, whereas our search is a deterministic enumeration of an audited 479-entry SMIRKS library gated by a chemistry-validity allowlist. Pocket2Mol(30) and DiffSBDD(31) are pocket-conditional generators that emit ligand atoms directly from a 3D pocket; they require a co-crystal structure and a pocket extraction step that our pipeline deliberately does not assume, trading the protein-side richness for the ability to operate on any lead for which a 2D ligand-interaction diagram is available. TamGen(32) is a target-aware chemical-language-model generator with reported sub-micromolar wet-lab hits against *Mycobacterium tuberculosis* ClpP, demonstrating that generative LLM-backed pipelines can reach experimentally validated leads; we share the LLM backbone and the target-aware framing but consume the target signal as a vision classification of an LID rather than as a learned pocket embedding. EquiBind(33) and DiffDock(34) are docking models rather than generators; they appear in our comparison only because the present pipeline can consume docking output as a numeric input to ranking but does not compute docking internally. PocketGen(35) inverts the problem (design the pocket given a ligand) and is therefore complementary rather than comparable.

For vision-language molecule understanding, the most directly related work is MolReGPT(36), GIT-Mol(37), MolFM(38), and DECIMER.ai(39). MolReGPT(36) demonstrates that in-context learning with retrieved exemplars lets an off-the-shelf ChatGPT model match fine-tuned MolT5 on molecule-caption translation; the lesson the present pipeline applies is the same (use the production model with a fixed structured prompt, do not fine-tune a per-task model), but our target is structured JSON over a labelled-instance list, not free-text captioning. GIT-Mol(37) and MolFM(38) integrate graph, image, and text modalities into a unified latent space and are evaluated on property prediction and molecule generation; their architectural ambition (one model, three modalities) is greater than ours, but the evaluation does not measure the specific quantity we care about (robustness of LID-derived restricted-atom classification), and they do not address downstream defence in depth at all. DECIMER.ai(39) is the closest specialised vision tool, an open-source EfficientNet-V2 plus Transformer pipeline that reads a printed 2D structure depiction and emits SMILES; it does not address binding-interaction extraction from LIDs, which is the specific vision sub-task we depend on. Across all of this work, the empirical gap the present paper addresses is not “LLMs can do chemistry” (already shown) and not “vision-language models can read molecule images” (already shown), but rather “how do you bolt an imperfect vision-language classifier into a production-grade analog generator without letting its perception errors corrupt the output set”, and the answer this paper documents is the defence-in-depth wrapping reported in §4.1.

A useful, current systematic survey of this fast-moving space is the Ramos et al. review of large language models and autonomous agents in chemistry(40), which catalogues thirty-plus chemistry-focused agentic systems published between 2023 and 2024 and observes that very few of them carry deterministic post-hoc guardrails; the present paper can be read as one concrete realisation of what those guardrails look like in production for lead optimization.

4.5 Failure-mode analysis

The defence-in-depth wrapping documented in §2 absorbs many but not all classes of error, and the failure surface that remains is worth making explicit. Four failure modes the architecture is most exposed to are: (i) vision-model false negatives, where the classifier misses a binding contact entirely and a restricted group is allowed into the editable set; (ii) SMIRKS library coverage gaps, of which the indole-to-quinoline ring-expansion case in §3.1 is the empirically surfaced exemplar but is not the only one (any ring-size-changing operation, any reagent-implicit transform such as N-alkylation under specific protection, and any stereochemistry-modifying transform are in the same scope-of-substitution-vocabulary class); (iii) soft Murcko gate over-permissiveness, where a ring-topology-preserving edit silently changes the electronics of the scaffold (for example, a phenyl-to-pyrimidine swap inside a ring system) and the gate’s predicate (same ring count, same aromaticity) does not see the electronic change; and (iv) chemistry-validator over-restrictiveness, where the static allowlist refuses a non-canonical but real interaction (e.g. a fluorine acting as a weak H-bond acceptor in a specific geometric context, or a sulfur participating in a non-standard cation- π interaction).

A fifth class the author should consider is structural-alert false positives, by which we mean PAINS or Brenk substructures that are legitimate medicinal-chemistry motifs and whose presence in the analog set is not a real liability. Capuzzi et al.(41) re-analysed the original PAINS-A/B/C filters against a 95,000-compound in-house pharmaceutical screening collection and reported that 97 % of PAINS-flagged compounds were infrequent rather than promiscuous hitters, and that 68 % of the 480 published alerts were derived from four or fewer compounds, so the alert-positive predictive value is target- and assay-class-dependent rather than universal. The Saubern et al. analysis(42) corroborated this on a separate pharmaceutical data set. In the present pipeline, the audit numbers in §3.2 show that the SMIRKS library raises the Brenk-flagged rate from 26.1 % on the seed compounds to 46.5 % under the gate-on default; some of that rise is real (the library introduces alert-substructure-bearing products) but some is artefactual structural-alert flagging in the sense Capuzzi documented. A practical mitigation is to (a) treat PAINS-A separately from PAINS-B/C (the latter two are derived from much smaller compound counts and are known to be noisier), (b) attach the alert as an informational tag with a soft score penalty rather than as a hard reject when the indication is target-class-known and the alert is on the lower-evidence list, and (c) cross-check alert-flagged analogs against a more recent assay-class-aware liability set (e.g. the Eli Lilly MedChem rules of Bruns and Watson(43)) before final reporting.

A sixth and arguably load-bearing failure mode is ADMET-prediction distribution shift. The Chemprop V2 backbone used by `admet_ai` (2) is trained on the Therapeutics Data Commons ADMET benchmark group(44), whose 22 datasets range from 475 to 13,130 molecules per endpoint with scaffold-split 80/20 holdouts. Recent out-of-distribution work(45) reports that ADMET models degrade substantially when test compounds occupy chemical space outside the training set’s scaffold neighbourhoods, and the SMIRKS engine in this pipeline is deliberately designed to take the lead off its scaffold; the very transformations the engine is built to perform push the output into regions of chemical space where the ADMET screen is least

calibrated. The present pipeline does not currently emit a per-endpoint applicability-domain confidence flag; adding one (e.g. by Tanimoto-similarity to the nearest training-set neighbour, or by the Mahalanobis distance in the Chemprop fingerprint space) would let the ranking step downweight predictions that are unreliable rather than treat all predictions as equally informative. A seventh failure mode, worth mentioning briefly, is synthetic-accessibility metric brittleness: SYBA(1) is a Bayesian classifier trained on a fixed easy-to-synthesise and hard-to-synthesise corpus and inherits that corpus's coverage limits; on highly novel scaffolds the score can be over-confident, and the present pipeline's centring-on-zero sigmoid penalty correctly softens this, but does not eliminate it. The Bender and Cortés-Ciriano(46) two-part review on what is realistic and what is illusion in AI for drug discovery is the right meta-reference for calibrating expectations on all of the above.

4.6 Regulatory and medicinal-chemistry implications

In the standard hit-to-lead and lead-optimization workflow, the output of a pipeline of this kind, namely a ranked list of pharmacophore-preserving, ADMET-screened, structural-alert-filtered analogs with provenance manifests, sits upstream of the medicinal chemist's bench prioritization and informs which analogs are synthesised, assayed in vitro for target binding and orthogonal selectivity, and progressed to animal PK. The most natural integration point is the SAR cycle: each iteration's wet-lab data updates the project-context analysis (stage 5), which re-weights the next round's ADMET screen and the ranker's endpoint priorities. None of the steps in this pipeline are binding-validated, in the sense the author explicitly notes in §1 and §4.2(d); the role of the pipeline is to *narrow* the candidate set the chemist examines, not to assert that any specific output binds the target. The pipeline's role in ADME-tox triage is more direct: because the per-analog ADMET vector is computed by the Chemprop V2-backed `admet_ai` (2) on the TDC benchmark group(44), the liability signals (hERG, AMES, DILI, CYP inhibition) are commensurable with the predictions the medicinal-chemistry community already uses to prioritize tox-axis follow-up, and they fall under the long-running in-silico-tox-prediction framework reviewed by Vamathevan et al.(47).

The regulatory picture is the right place to be conservative. For genotoxicity and mutagenicity specifically, ICH M7(R2)(48) is the operative international guideline and is the first regulatory document to formally accept in-silico (Q)SAR results in place of in-vitro Ames testing for initial impurity-hazard classification, requiring two complementary methodologies (a rule-based and a statistical model) followed by expert review. The pipeline's structural-alert filter (PAINS, Brenk) is *not* a substitute for the ICH M7 (Q)SAR step; PAINS-A/B/C are pan-assay interference filters, Brenk is a neglected-disease lead-discovery filter, and neither is the validated genotox classifier ICH M7 contemplates (e.g. Derek Nexus and Sarah Nexus, or Leadscope and CASE Ultra), but the pipeline's output is a natural input to that downstream step. On the broader AI/ML-in-drug-development front, the U.S. FDA's January 2025 draft guidance "Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products"(49) sets out a risk-based credibility-assessment framework with seven enumerated steps (define the question of interest, define the context of use, assess the risk, develop a credibility-assessment plan, execute the plan, document the results, determine adequacy), and the European Medicines Agency's September 2024

reflection paper(50) sets out broadly aligned expectations across the EU. Both frameworks place heavy emphasis on **provenance, reproducibility, and applicability-domain documentation**, which is precisely what this pipeline’s per-experiment `manifest.json` (git SHA, RDKit version, Python version, SMIRKS library SHA-256, input dataset SHA-256) is engineered to support. The cross-provider model benchmark in §3.5 also contributes here: documenting that the production model and the fallback chain were each evaluated against the same prompt with the same rubric is exactly the kind of credibility evidence the FDA’s seven-step framework asks for.

4.7 Bias and reproducibility risks

Three classes of bias and reproducibility risk are baked into a pipeline of this shape and deserve explicit naming. First, **training-data bias in the ADMET backbone**: `admet_ai(2)` wraps Chemprop V2(3) message-passing networks (extending the directed-message-passing architecture introduced by Yang et al.(4)) trained on the TDC ADMET benchmark group(44), which is itself an aggregation of public single-task data sets (e.g. AqSolDB for solubility, Caco-2 from Wang et al., Lipophilicity from ChEMBL, hERG from Karim et al., AMES from Hansen et al.). Each underlying assay has its own selection bias on chemical space (literature-reported compounds are not a uniform sample of drug-like space; positive examples are over-published; certain laboratory standard scaffolds, kinase hinge binders, GPCR amine fragments, are over-represented), and recent OOD-benchmark work(45) shows that scaffold-shifted test sets degrade ADMET predictions substantially. The practical consequence for our pipeline is that the ADMET screen at stage 10 is most reliable for analogs that stay close to the lead’s scaffold, and least reliable for analogs the SMIRKS engine pushes furthest from it, which is the opposite of what one would naively hope.

Second, **vision-language model training-data origin** is essentially opaque for the production-class models in use. The vendors do not publish LID-image-specific training-corpus inventories, and there is no public audit of how many ligand-interaction-diagram images each model has seen, drawn from which sources (LigPlot+, PoseView, PLIP, manually drawn in patents, manually drawn in journal figures), at what resolution, in which colour scheme, or with which residue-label convention. The self-consistency Jaccard of 1.00 reported in §3.3 measures *reproducibility on a single LID*; it does not measure accuracy on out-of-distribution LIDs drawn from a different rendering pipeline (for example, a Maestro-rendered LID with a different arrow-head convention than the PoseView convention used in our test set). The closest related quantification of vision-model behaviour drift is the LLM-output-drift work reported by Khatchadourian and Franco(51) and the original ChatGPT-behaviour-changing-over-time study by Chen, Zaharia, and Zou(52), both of which document substantial month-over-month variation in fixed-prompt outputs from frozen-version model APIs.

Third, **provider drift and model deprecation** is the practical reproducibility nightmare for any LLM-backed production pipeline. Chen et al.(52) measured GPT-3.5 and GPT-4 on a fixed set of tasks across three months and found that GPT-4’s prime-vs-composite accuracy fell from 84 % to 51 % between the March and June 2023 versions of the *same* named API endpoint; code-generation formatting accuracy degraded over the same window; instruction-following

ability fell. Khatchadourian and Franco(51) extend this analysis with a cross-provider validation framework and document that even when the model name and version pin are held constant, vendor-side fine-tuning, safety-filter retuning, and infrastructure-level sampling-default changes can shift output distributions in ways that break downstream consumers. The mitigations the present paper implements against all three of these risks are concrete and worth restating: (a) every experiment carries a `manifest.json` recording the git SHA, RDKit version, Python version, platform, SMIRKS library SHA-256, and input dataset SHA-256, so the deterministic part of the pipeline can be reproduced to the byte; (b) structured-output flags (`response_format: json_object` for OpenAI and Mistral, `response_mime_type: application/json` for Gemini) are enabled explicitly per §2.21 Methods and §3.5 Results, so cross-provider portability of the JSON-shape contract is load-bearing rather than implicit; (c) HTTP-429 and HTTP-503 retries with exponential backoff (delays 2, 4, 8, 16 s; up to five attempts) ensure transient provider-side unreliability does not corrupt benchmark results; and (d) the cross-provider model benchmark itself, evaluated against eleven text and fourteen vision models with the production prompts verbatim, is the documentation: it tells the next person to reproduce or audit the pipeline which model produced which behaviour at which date, so that a future provider-side regression can be detected by re-running the benchmark on the same inputs. None of this eliminates provider drift; all of it makes provider drift detectable and recoverable rather than silent.

5. Conclusions

We have presented a twelve-stage agentic pipeline for lead optimisation that combines vision-language ligand-interaction-diagram parsing with deterministic chemistry-validity gates, a 479-entry SMIRKS transformation library, upstream ADMET prediction via the Chemprop V2 backbone, and SYBA-anchored synthetic-accessibility ranking. The architectural contribution is defence-in-depth: each language-model agent stage is wrapped by a deterministic gate so that perception errors absorbed at one stage cannot corrupt the analog set downstream. Across five evaluations we have shown that the SMIRKS engine exact-recovers 53.3 % of curated single-edit bioisosteric pairs and 15.5 % of unbiased ChEMBL-37-derived pairs (mean best Tanimoto-to-target 0.755 in misses), that the soft Murcko gate cuts the analog set roughly in half while raising scaffold preservation to 100 %, that the vision-language classifier exhibits perfect pairwise Jaccard self-consistency on a fixed input with zero chemistry-validator drops, that a LID-aware end-to-end run on a DYRK1A kinase / CNS lead improves the rank-1 analog on both Pareto-style score and synthetic-accessibility relative to a LID-free baseline, and that a cross-provider benchmark across eleven language models identifies reliable configurations for the production default.

The platform is implemented as production software and the code, data, manifest-stamped raw outputs, and reproduction scripts are released open source under the MIT licence at <https://github.com/Nimieeee/lead-optimizer-paper> with a frozen Zenodo archive at <https://doi.org/10.5281/zenodo.20643485>. We invite the cheminformatics community to reuse the pipeline, to contribute additional SMIRKS transformations that close the documented coverage gaps such as ring-size-changing operations, and to extend the evaluation to held-out gold-

standard ligand-interaction-diagram sets so that the vision-language classifier’s accuracy (not only its self-consistency) can be characterised across diverse target classes.

Methods Availability

All scripts used in this study, including the five experiment drivers, the figure-rendering scripts, the cross-provider model-benchmark harness, and the mirrored pipeline source code (`code/lead_optimizer/`), are available at <https://github.com/Nimieeee/lead-optimizer-paper>. The exact commit used to produce every figure and table in this manuscript is archived on Zenodo at <https://doi.org/10.5281/zenodo.20643485> (release `v1.0-chemrxiv`); that Zenodo DOI is the canonical citation for the artefact. The repository carries a `manifest.json` for every experiment recording the git SHA, RDKit version, Python version, platform, SMIRKS library SHA-256, and input-file SHA-256, so that any figure or table can be re-derived to the byte.

Use of Artificial Intelligence Tools

Portions of this manuscript were prepared with the assistance of Anthropic’s Claude (Opus 4.8). The AI assistant was used to support iterative drafting, clarity refinement, editorial polishing, figure-caption generation, the writing of experiment driver scripts (Python / RDKit / matplotlib) and figure-rendering scripts, and orchestration of the benchmark harness across providers. All scientific content, experimental design, claim verification against measured results, interpretation of findings, and final conclusions, was conceived, executed, and validated by the author. The AI assistant was employed as an editorial, computational-scripting, and orchestration aid and was not used to generate novel scientific claims or to perform data interpretation.

Data Availability

The matched-molecular-pair pilot set (30 hand-curated single-edit pairs), the 129-marketed-drug seed list, the 50 LID image used for the vision self-consistency study, the four representative project-context strings, and the four leads used in the cross-provider model benchmark are all included in the repository under `experiments/exp*/data/`. The unbiased ChEMBL-derived MMP set was built from ChEMBL release 37 (chemreps file), a 2,000-compound drug-like subset filtered with the criteria reported in §3.1, then fragmented and indexed with `mmpdb` to produce 41,310 raw pairs of which 2,000 single-edit pairs were sampled for evaluation; the build script (`experiments/exp1_mmp_recovery/build_chembl_pairs.py`) is in the repository and the raw ChEMBL chemreps file is excluded only because of its size and is downloadable from EBI at <https://ftp.ebi.ac.uk/pub/databases/chembl/>. All experiment-result JSONs and per-run manifests are committed in `experiments/exp*/results/`. A frozen archive of the exact commit used to produce every figure and table in this manuscript is deposited on Zenodo and assigned a DOI; that DOI is the canonical citation for the data artefact.

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